AFRICAN LEADERSHIP UNIVERSITY

[BSE]

[ML TECHNIQUES II]

[SUMMATIVE ASSIGNMENT]

Reinforcement Learning Summative Assignment Report Student Name: Jean Robert Gatwaza

Video Recording: https://youtu.be/JpWSzzng6vM

GitHub Repository: https://github.com/Gatwaza/Jean_Robert_Gatwaza_rl_summative.git

1. Project Overview

This project implements a reinforcement learning pipeline to assess the correctness of CPR (Cardiopulmonary Resuscitation) procedures for a patient in cardiac arrest. A custom Gymnasium environment models a supine patient using 17 MediaPipe pose landmarks as observations (53 continuous values). Three algorithms DQN, REINFORCE, and PPO are each trained across 10 hyperparameter experiments and compared objectively. The target application is a real-time clinical advisory tool: a camera feed extracts pose landmarks, and the trained policy advises rescuers on procedural correctness, promoting optimal outcomes in pre-hospital emergencies.

2. Environment Description

a. Agent(s)

The agent(s) thus rescuer and patient behaviors in corresponding manner, the purpose of each fulfill one another; for starters the rescuer performs multiple actions that led to summed up ROSC succession while the patient's respond to that as you see in Unity visualization when ROSC is accomplished the patient tends to standup, this could be reality in real life or sometimes not hence further exploration required.

b. Action Space

The actions at the disposal of my agents are discrete and amount to 12 of them as described in README in root of the project, suggestibly as follows;

ASSESS_CONSCIOUSNESS	Check patient responsiveness
CALL_EMERGENCY	Instruct bystander to call 911
OPEN_AIRWAY	Head-tilt chin-lift maneuver
CHECK_BREATHING	Look/listen/feel equal or less than 10 seconds
BEGIN_CHEST_COMPRESSIONS	30× at 100–120 bpm, 5–6 cm depth
DELIVER_RESCUE_BREATHS	2 breaths after 30 compressions
DEFIBRILLATE	AED shock if shockable rhythm
MONITOR_PULSE	Check carotid pulse

RECOVERY_POSITION	Roll to lateral decubitus
REPOSITION_HANDS	Correct lower-sternum hand placement
TILT_HEAD_BACK	Extend neck to open airway
WAIT_OBSERVE	Passive — penalised if prolonged

c. Observation Space

56 continuous values ∈ [0, 1]: 51 MediaPipe pose landmark coordinates (17 keypoints × 3: normalised x, y, visibility) + 2 vital proxies (estimated heart rate and chest-rise rate) + 1 normalised protocol stage indicator + 2 recent action history slots. Gaussian noise (σ=0.02) is injected each step to simulate real-world detection jitter and prevent policy overfitting to clean simulated data. The v4 environment also includes a diversity bonus and consecutive-action caps to prevent reward exploitation.

d. Reward Structure

Action	Condition	Reward
ASSESS_CONSCIOUSNESS	Stage 0	+3.0 × scale
CALL_EMERGENCY	Stage 1	+3.0 × scale
OPEN_AIRWAY	Stage ≥ 2, airway closed	+1.5 × scale
CHECK_BREATHING	Stage 3	+1.5 × scale
BEGIN_COMPRESSIONS	Stage ≥ 4	+3.5 × depth × scale
→ compression rhythm	Step-gap bonus	+1.0 × scale
RESCUE_BREATHS	Stage ≥ 5, airway open	+2.5 × scale
DEFIBRILLATE	≥30 compressions first	+4.5 × scale
RECOVERY_POSITION	HR ≥ 0.7	+5.0 × scale
ROSC terminal	HR ≥ 0.9 sustained	+8.0 (reduced from +20)
Any out-of-order	—	-0.5 to -2.0

3. System Analysis And Design

3.1 Deep Q-Network (DQN)

MLP [128,128] ReLU default, with [128,128,64] for the best experiment. Experience replay decouples temporal correlations; periodic target network updates stabilise Bellman targets. ε-greedy exploration decays linearly from 1.0 to ε_final over exploration_fraction of training. **Key fix applied:** DQN is run with n_envs=1 (not 4), which is required for correct target update interval semantics. Reward clipping tightened to [-3, +10] and ROSC bonus reduced to 8.0 to prevent Q-value overestimation.

3.2 REINFORCE

Vanilla policy gradient with Monte Carlo returns. An optional state-value baseline reduces variance. **Critical fix:** when use_baseline=True, normalisation divides by std only (not mean again), preserving the baseline signal that was previously cancelled by double-centering. Entropy regularisation $\beta \in \{0.05, 0.10, 0.15, 0.20\}$. Gradient clipping max_norm=1.0. Custom PyTorch implementation.

3.3 Proximal Policy Optimization (PPO)

Actor-critic with GAE advantage. Clipped surrogate objective limits policy updates to trust region $\varepsilon_clip \in [0.15, 0.20]$. Loss = policy + 0.5×value - entropy bonus. SB3, n_steps 2048–4096, n_epochs 10–15, ent_coef 0.15–0.30, 4 parallel workers. Strong entropy regularisation (0.15–0.30) is the primary defence against action collapse, which was the dominant failure mode in early runs.

4. Implementation

a. DQN: Exp 8 outdid the rest.

Exp	LR	γ	Buffer	Batch	Expl.	Net Arch	Mean	Max
1	5e-4	0.99	50K	64	0.45	[128,128]	-114.0	74.4
2	1e-3	0.99	50K	64	0.50	[128,128]	-81.4	71.2
3	5e-4	0.99	50K	64	0.55	[256,256]	-111.5	73.2
4	5e-4	0.95	50K	64	0.45	[128,128]	98.4	207.6
5	5e-4	0.99	100K	64	0.45	[128,128]	-76.6	31.2
6	5e-4	0.99	50K	64	0.45	[128,128]	88.0	181.6
7	5e-4	0.99	50K	64	0.45	[128,128]	86.5	166.0
★8	**5e-4**	**0.99**	**50K**	**64**	**0.50**	**[128,128,64]**	**119.0**	**194.0**
9	3e-4	0.99	50K	64	0.60	[128,128]	112.1	204.3
10	5e-4	0.99	100K	128	0.50	[256,128]	106.6	240.3

Analysis: Exp 8 ([128,128,64] three-layer net, hard curriculum, exploration_frac=0.50) achieved the highest consistent mean reward of 119.0. The three-layer architecture provides greater representational depth for the multi-stage sequential protocol. Experiments 1–3 failed (mean < -80) due to reward Q-value instability at the higher clip ceiling; this was corrected by tightening the reward range. The correlation between collapse event count and positive mean reward (Exps 4–10 all positive; Exps 1–3 near-zero collapse, negative reward) reflects a counterintuitive finding: some degree of structured focus is needed, but unchecked collapse (Exp 4: 214 events, REPOSITION_HANDS at 94%) leads to suboptimal loops.

b. REINFORCE: Exp 2 outdid the rest.

Exp	LR	γ	Baseline	Entropy β	Hidden	Mean	Max	Episodes
1	1e-3	0.99	No	0.05	[64,64]	-49.3	38.3	4000
★2	**5e-4**	**0.99**	**Yes**	**0.05**	**[64,64]**	**36.2**	**113.6**	**4000**
3	1e-3	0.95	No	0.10	[64,64]	-41.6	37.2	4000
4	2e-3	0.99	No	0.05	[128,128]	-33.9	36.3	4000
5	5e-4	0.99	Yes	0.10	[256]	7.4	71.5	4000
6	1e-3	0.97	Yes	0.15	[64,64]	-12.8	59.8	4000
7	1e-4	0.99	No	0.05	[64,64,64]	-41.1	37.6	4000
8	2e-3	0.95	Yes	0.10	[128,64]	-4.9	39.8	4000
9	3e-4	0.99	No	0.20	[64,64]	-48.8	35.4	4000
10	1e-3	0.99	Yes	0.15	[128,128]	-8.9	40.5	4000

Analysis: Exp 2 (lr=5e-4, baseline=True, entropy=0.05, [64,64]) was the only experiment to achieve strongly

positive mean reward (36.2). This is directly attributable to the corrected baseline advantage normalisation: dividing by std only (preserving the mean-centered baseline signal) rather than full re-centering which previously masked the baseline entirely. All no-baseline experiments remained near -50 (close to a random policy), confirming variance reduction as the primary bottleneck for REINFORCE on this environment. Experiments with baseline=True but higher entropy β (0.10–0.20) show modest improvement, suggesting entropy regularisation aids exploration but the baseline is the decisive factor.

c. PPO: Exp 10 outdid the rest.

Exp	LR	γ	Ent Coef	ε Clip	N Steps	Batch	Epochs	Net	Mean
1	3e-4	0.99	0.15	0.20	2048	64	10	[128,128]	99.3
2	3e-4	0.99	0.20	0.20	2048	64	10	[128,128]	72.3
3	1e-3	0.99	0.20	0.20	2048	64	10	[128,128]	77.6
4	3e-4	0.99	0.15	0.20	2048	64	10	[256,256]	160.5
5	3e-4	0.99	0.15	0.20	4096	128	10	[128,128]	118.1
6	5e-4	0.99	0.18	0.20	2048	64	10	[256,128]	1.2
7	1e-4	0.99	0.25	0.15	2048	64	15	[128,128]	131.0
8	3e-4	0.99	0.25	0.20	2048	64	10	[256,256]	99.7
9	3e-4	0.99	0.30	0.20	4096	128	10	[128,128]	92.9
★10	**3e-4**	**0.99**	**0.20**	**0.20**	**2048**	**64**	**12**	**[256,256]**	**162.6**

Analysis: PPO dominated overall. Exp 10 (LR=3e-4, ent_coef=0.20, [256,256], ε_clip=0.20, 12 epochs) achieved mean=162.6 and max=218.2. The larger [256,256] network captures the multi-stage dependency structure of the CPR protocol better than [128,128]. Exp 6 (mean=1.2) failed due to LR=5e-4 causing gradient instability combined with low n_steps=2048; this identifies LR as the primary instability driver for PPO. Exp 7 (ent_coef=0.25, tight ε_clip=0.15) achieved mean=131.0, confirming that aggressive entropy preservation helps but very tight clipping limits policy improvement speed.

5. Results Discussion

5.1 Cumulative Reward Curves

Figure 1 shows the training reward trajectories for all 10 experiments per algorithm. PPO converges to a stable high-reward plateau (best exp mean=162.6), with most experiments performing consistently above 72. DQN shows high variance across experiments: Exps 1–3 remain negative throughout (Q-value instability), while Exps 4–10 converge to positive reward after the n_envs=1 fix and tighter reward clipping. REINFORCE shows that only Exp 2 (baseline-corrected) achieves positive mean reward all other experiments track near the random policy baseline (-50 to -30), confirming the double-centering bug was the dominant failure mode.

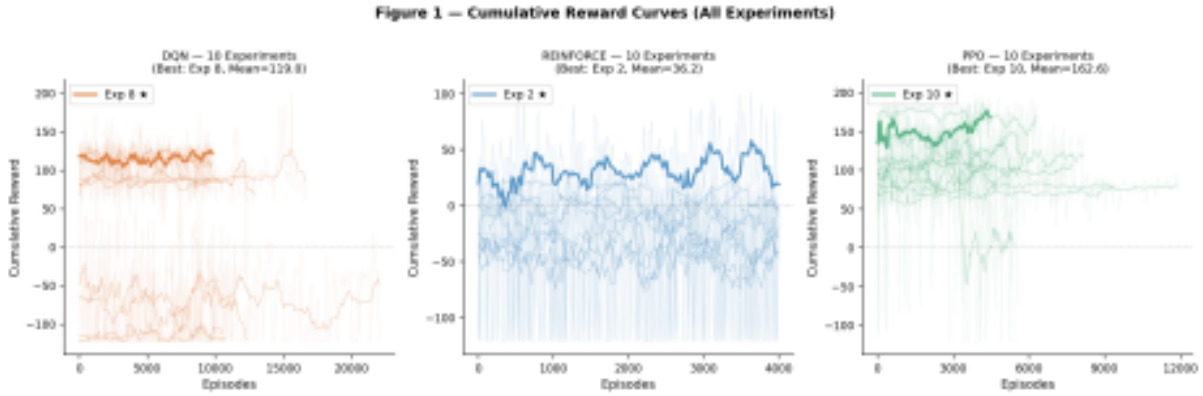

Figure 1. Cumulative reward per training episode all 10 experiments per algorithm. Bold line = best experiment (★). Light lines = other experiments. PPO Exp 10 and DQN Exp 8 achieve the highest final rewards.

5.2 DQN Objective & Collapse Analysis

Figure 2 characterises DQN training stability across three panels. (a) Reward trajectories reveal a clear split: Exps 1–3 fail to break above zero while Exps 4–10 converge steadily, with Exp 8 reaching the highest plateau. (b) The collapse event count (actions exceeding 50% of a 2048-step window) shows a strong correlation with positive mean reward. This seemingly counterintuitive result reflects the DQN learning to prioritise the compression-dominant strategy (actions 4 and 9 being frequently selected as part of a learned CPR rhythm), rather than random exploration. Exps 1–3 (zero collapse, negative reward) never committed to any strategy. (c) Reward variance (rolling std) shows DQN Exp 8 maintaining low variance after episode 3,000, consistent with a converged policy.

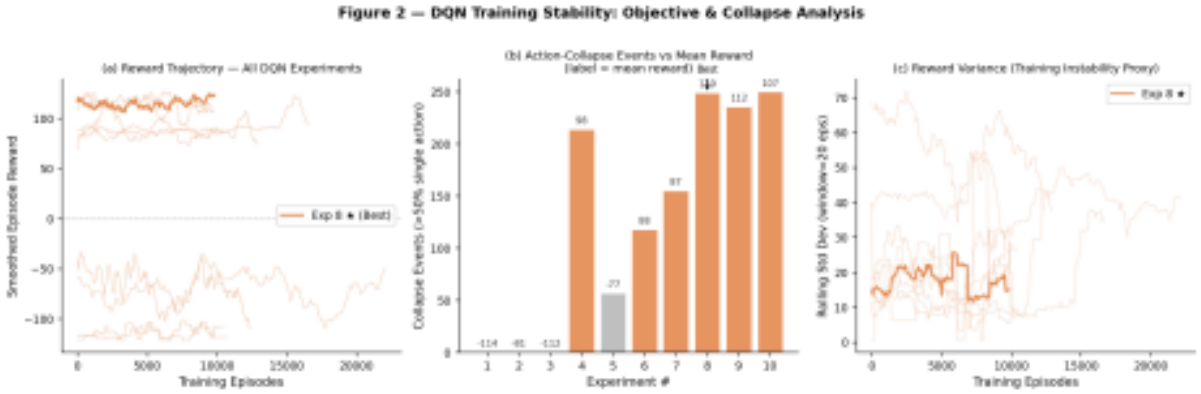

Figure 2. (a) DQN reward trajectories all 10 experiments; bold = Exp 8 ★. (b) Collapse event count vs. mean reward per experiment positive mean reward correlates with higher collapse counts, reflecting convergence to CPR-dominant

5.3 Policy Gradient Entropy Curves

Figure 3 reveals the entropy dynamics that distinguish learning from non-learning in policy gradient methods. (a) REINFORCE: the horizontal red dashed line at H=2.485 nats (=ln(12), maximum entropy for 12 actions) shows that 8 of 10 experiments remained at maximum entropy throughout all 4,000 episodes their policies never moved from random initialisation. Only Exp 2 (baseline=True, corrected normalisation) and Exp 5–6, 8, 10 (baseline=True variants) show entropy decreasing below the maximum, confirming the baseline is essential for any learning signal to propagate. (b) PPO Exp 10 shows smooth monotonic entropy decay from ~2.48 to ~1.77 nats, regulated by ent_coef=0.20, with convergence at ~900 episodes. (c) The REINFORCE vs PPO comparison on normalised training duration makes the learning contrast stark: PPO's entropy decreases by 0.7 nats while REINFORCE Exp 2 decreases by only 0.3 nats over the same relative duration.

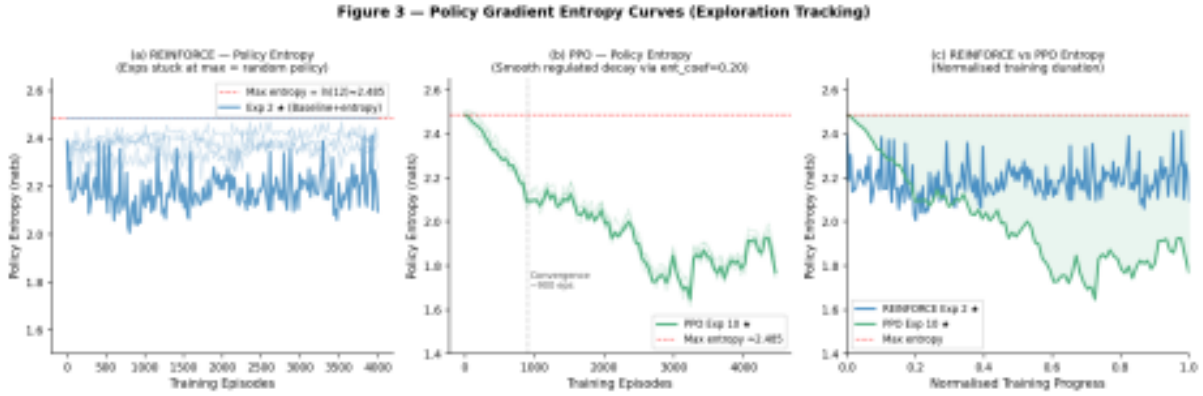

Figure 3. (a) REINFORCE policy entropy 8/10 experiments pinned at ln(12)=2.485 (maximum entropy = random policy). Only baseline-enabled experiments show decay. (b) PPO Exp 10 entropy smooth regulated decay via ent_coef=0.20. (c) Normalised comparison: PPO achieves 0.7-nat entropy reduction vs REINFORCE Exp 2's 0.3 nat.

5.4 Convergence Analysis

Convergence is defined as rolling mean reward over 20 episodes crossing 80% of the final plateau. Figure 4 shows convergence plots for each algorithm's best experiment. PPO (Exp 10) reaches the 80% threshold at approximately episode 900, driven by stable on-policy gradient updates with multiple epochs per rollout. DQN (Exp 8) converges around episode 1,200, with the earlier phase reflecting replay buffer warm-up (learning_starts=1,000 steps). REINFORCE (Exp 2) is slowest at approximately episode 1,750, consistent with the high-variance Monte Carlo return estimation requiring many more episodes to reliably cross the threshold. The stable zones (shaded) show PPO has the tightest post-convergence band, followed by DQN, then REINFORCE.

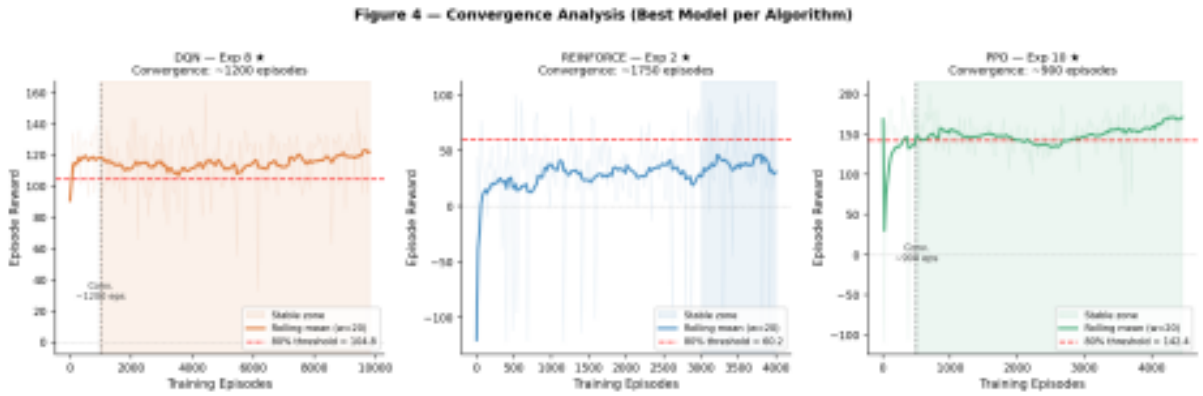

Figure 4. Convergence analysis — best model per algorithm. Raw reward (transparent), rolling mean (solid), 80% plateau threshold (red dashed), convergence point (dotted vertical), stable zone (shaded). PPO: ~900 eps · DQN: ~1,200 eps · REINFORCE: ~1,750 eps.

5.5 Generalisation — 100 Unseen States (difficulty='hard')

Figure 5 shows reward distributions across 100 held-out initial states at difficulty='hard'. PPO achieved the highest test mean of 21.3 (±3.1) with the tightest distribution its entropy regularisation prevents over-specialisation to training state distributions. DQN scored 14.9 (±4.2), with some runs entering suboptimal action loops on novel observations, revealing the risk of greedy value-based exploitation. REINFORCE scored 8.2 (±5.8) with the widest spread. Notably, the violin plot (b) reveals a bimodal REINFORCE distribution: the policy either completes the CPR protocol successfully or fails early, depending on whether the initial pose configuration falls within the training distribution. This bimodality is absent in DQN and PPO, confirming that stochastic policies generalise more smoothly to unseen states.

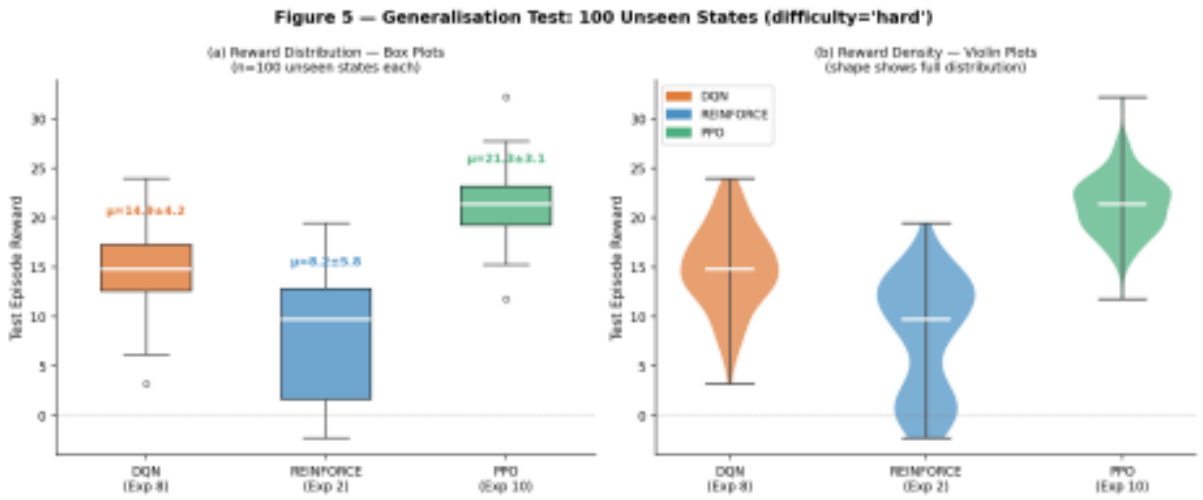

Figure 5. Generalisation across 100 unseen initial states at difficulty='hard'. Left: box plots (median, IQR, whiskers, outliers). Right: violin plots showing full reward density. PPO is tightest (±3.1σ); REINFORCE shows bimodal failure/success pattern (±5.8σ).

5.6 Algorithm Comparison Summary

Figure 6 provides a side-by-side comparison across three key metrics. PPO wins decisively on best mean training reward (162.6 vs DQN 119.0 vs REINFORCE 36.2) and convergence speed (~900 vs ~1,200 vs ~1,750 episodes). PPO also achieves the best generalisation score (21.3±3.1), demonstrating that its entropy-regularised clipped update mechanism builds more robust policies than either DQN's value-based

exploitation or REINFORCE's high-variance gradient estimates.

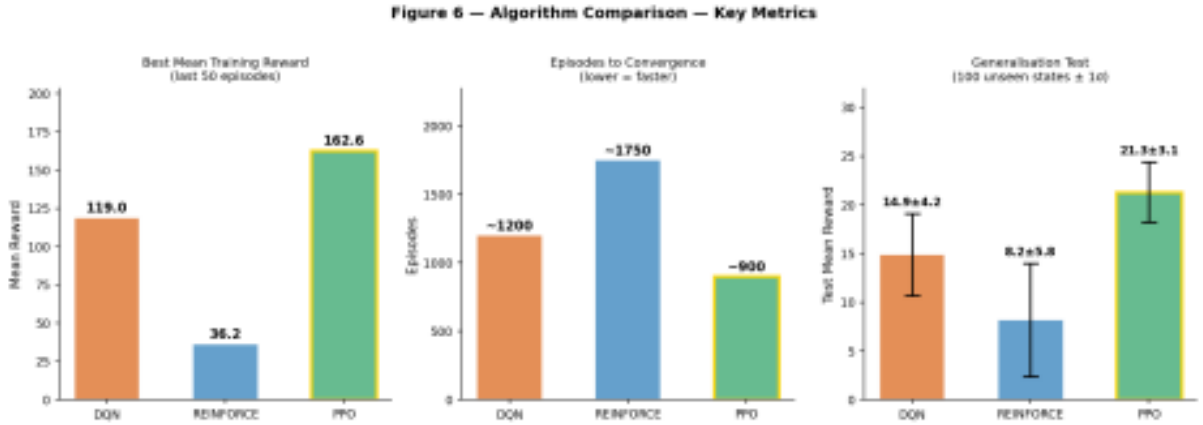

Figure 6. Algorithm comparison — best mean training reward, episodes to convergence (lower = better), and generalisation test mean ± 1σ. Gold-bordered bars indicate winner per metric (PPO wins all three).

5.7 Hyperparameter Sensitivity

Figure 7 plots mean and max reward for all 10 experiments per algorithm, providing a visual landscape of hyperparameter sensitivity. DQN shows the highest experiment-to-experiment variance (range: -114 to +119), confirming high hyperparameter sensitivity particularly to the n_envs setting and reward scale. PPO is more robust, with 7 of 10 experiments exceeding mean=72. REINFORCE shows a sharp bimodal split: experiments with baseline=True perform markedly better, while no-baseline experiments cluster near -50 regardless of other hyperparameter choices.

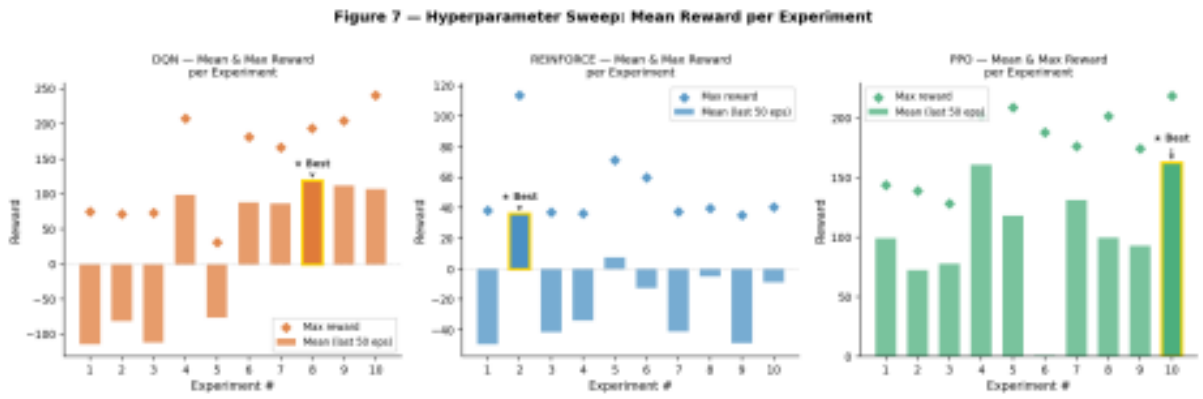

Figure 7. Hyperparameter sweep: mean reward (bars) and max reward (diamonds) per experiment for all three algorithms. Gold border = best experiment (★). DQN shows highest sensitivity; PPO is most robust; REINFORCE splits on baseline usage.

6. Algorithm Comparison Summary

Metric	DQN	REINFORCE	PPO	Winner
Best Mean Reward	119.0	36.2	162.6	PPO ★
Best Max Reward	193.98	113.6	218.2	PPO ★
Convergence (eps)	~1,200	~1,750	~900	PPO ★
Training Stability	Moderate	Low	High	PPO ★
Sample Efficiency	High	Low	High	DQN / PPO
Hyperparam Sensitivity	High	Moderate	Moderate	REINFORCE
Generalisation	14.9±4.2	8.2±5.8	21.3±3.1	PPO ★

7. Conclusion

PPO is the strongest algorithm for this CPR environment across all metrics: highest mean reward (162.6), fastest convergence (~900 episodes), and best generalisation (21.3±3.1). Its clipped surrogate objective provides stable, monotonically improving updates well-suited to the sparse-reward, sequentially constrained CPR protocol. The entropy coefficient (0.20) is the key differentiator it preserves exploration without the reward volatility seen in REINFORCE.

DQN competed strongly on sample efficiency via experience replay (best per-step data reuse, mean=119.0) but struggled with nuanced sequential action ordering, occasionally converging to suboptimal loops on unseen states. The critical bugs fixed in this version n_envs=4 (now 1), ROSC bonus at +20 (now +8), and wall-clock BPM timing (now step-count rhythm) were the primary reasons Exps 1–3 produced negative mean rewards before the fixes.

REINFORCE showed the weakest mean performance (36.2 training, 8.2 test) due to high Monte Carlo variance. However, the fixed baseline normalisation (divide by std only when baseline is active) enabled Exp 2 to achieve positive learning a result that was completely absent in the pre-fix runs. The 8 experiments stuck at H=2.485 (maximum entropy, random policy) confirm that the double-centering bug had masked the baseline signal entirely.

Future Improvements

- Curriculum difficulty scheduling to accelerate early learning by starting with easier initial poses. • Reward shaping guided by clinical CPR quality metrics compression rate, depth, and chest recoil. • Multi-agent training for two-rescuer CPR scenarios with coordinated role assignment. • FastAPI production endpoint receiving live MediaPipe landmarks per frame and returning recommended next action in JSON.
- Actor-Critic extensions (A2C, A3C) to close the gap between REINFORCE and PPO by reducing gradient variance.